41891: Cloud Computing Infrastructure
42891: Infrastructure for Cloud Computing

Week 3: Infrastructure/Management Mechanisms and Architectures

Contents

Infrastructure of Cloud Service Provider

* Broadband Networks and Internet Architecture
* Data Center Technology
* (Modern) Virtualization Technology
* Web Technology
* Multitenant Technology
* Service Technology

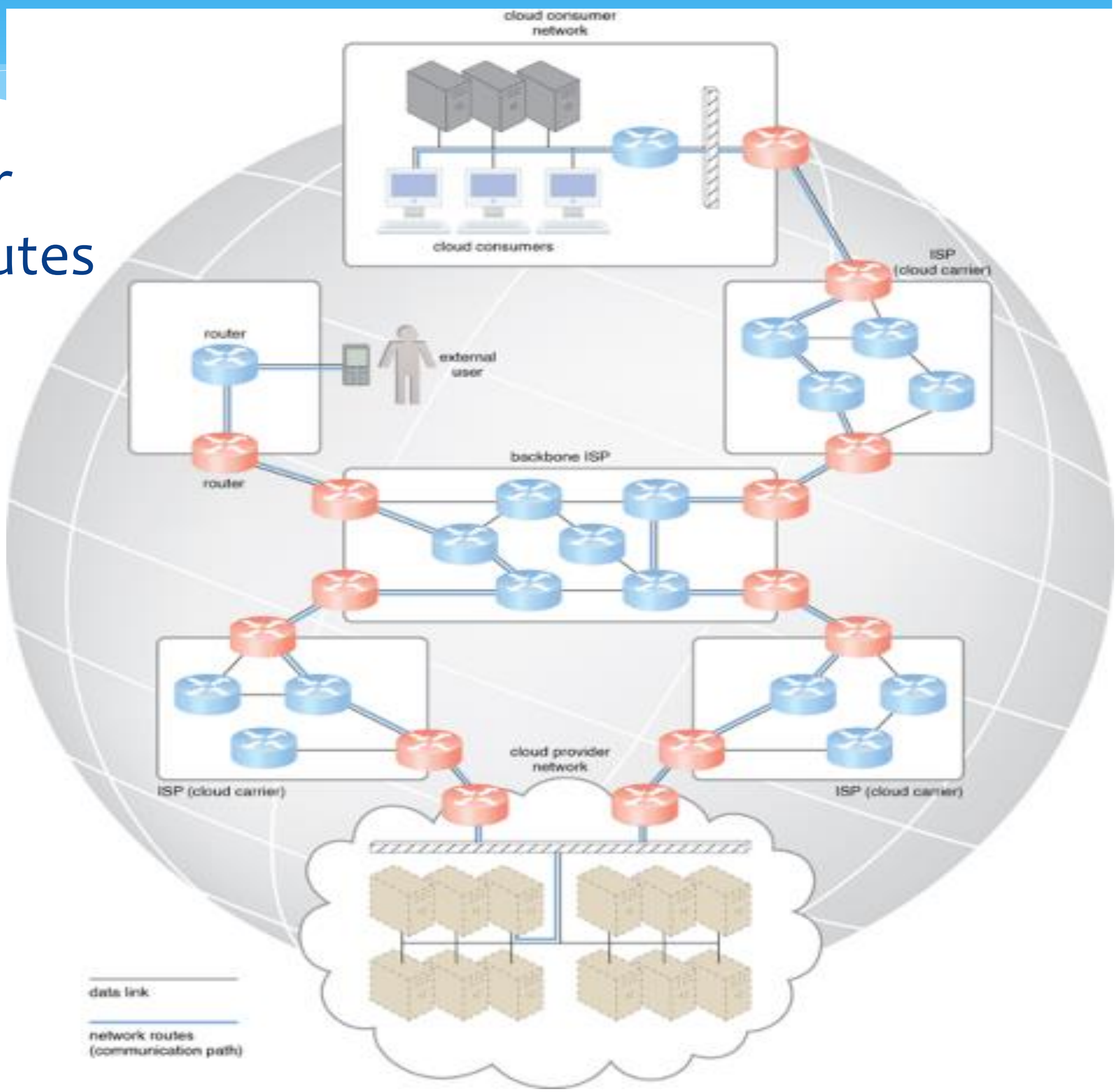
http://whatiscloud.com/origins_and_influences/technology_innovations

1. Broadband Networks and Internet Architecture

* Internet Service Providers (ISPs)
  * An ISP network interconnects to other ISP networks and various organizations
* Connectionless Packet Switching (Datagram Networks)
  * End-to-end data flows are divided into packets of limited size
* Router-Based Interconnectivity
  * Packets travelling through the Internet are directed by a router that arranges them into a message

Internet Service Providers (ISPs)

* Messages travel over dynamic network routes in this ISP internetworking configuration



2. Data Center Technology

* Facilities
  * Housing, racks, cabling, power supplies, environmental control station (heating, ventilation, air conditioning, fire protection, and other sub systems)
* Computing Hardware
  * Server, blade servers, etc.
* Storage Hardware
  * Hard Disk Arrays, Storage Area Network (SAN), Network-Attached Storage (NAS)
* Network Hardware
  * Carrier and External Network Interconnection, Web-Tier Load Balancing and Acceleration, LAN Fabric, SAN Fabric, NAS Gateways

3. Virtualization Technology

* Servers
  * Physical server can be abstracted into a virtual server
* Storage
  * Physical storage can be abstracted into a virtual storage device or virtual disk
* Network
  * Physical routers and switches can be abstracted into a logical network fabrics, such as VLANs
* Power
  * Physical UPS can be abstracted into a virtual UPS

Book: **Cloud Computing: Concepts, Technology & Architecture**. by Zaigham Mahmood, Ricardo Puttini, Thomas Erl. Released May 2013. Publisher(s): Pearson.

4. Web Technology and Service Technology

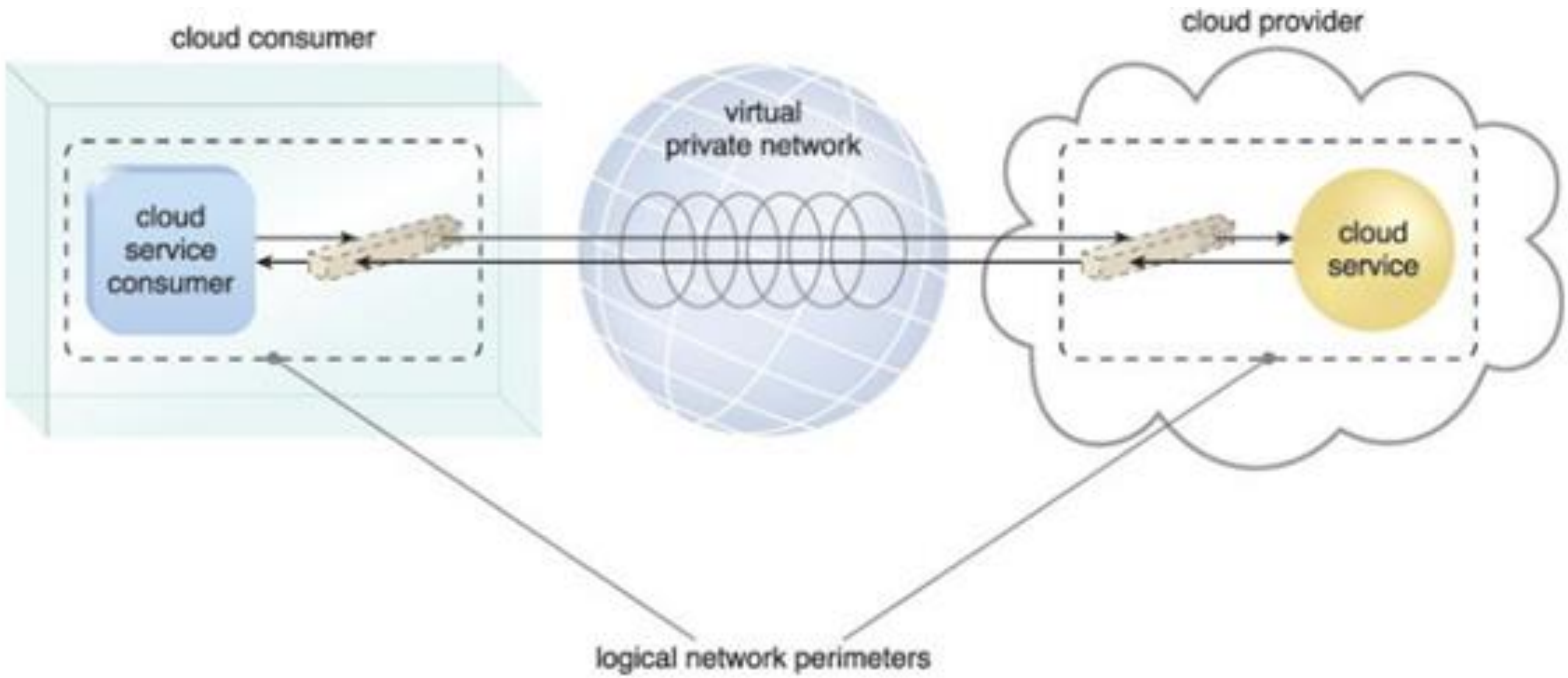
* Web Technology
  * Generally used as both implementation medium and the management interface for cloud services
* Service Technology
  * Keystone foundation of cloud computing
  * Web services, WSDL, XML schema, SOAP, UDDI, REST services, service agents, service middleware

Cloud Infrastructure Mechanisms

* Logical Network Perimeter
  * Isolation of a network from the rest of a communications network
* Virtual Server
  * Form of virtualization software that emulates physical server
* Cloud Storage Device
  * Represent storage devices that are designed specifically for cloud-based provisioning
* Cloud Usage Monitor
  * Autonomous software program responsible for collecting and processing IT resource usage data
* Resource Replication
  * Creation of multiple instances of the same IT resource
* Ready-made Environment
  * Defining component of PaaS cloud delivery model equipped with a complete software development kit

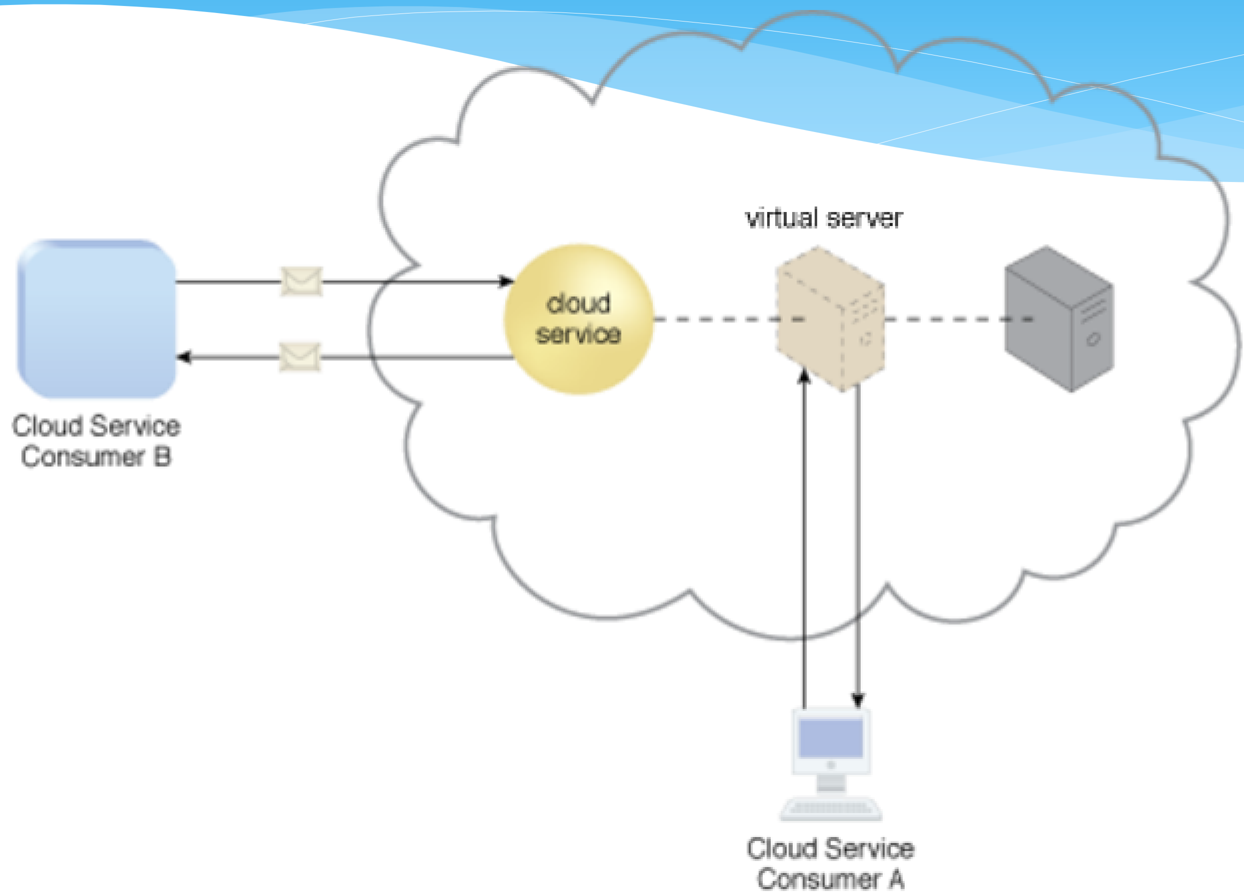
Cloud computing patterns, mechanisms: http://cloudpatterns.org/mechanisms/overview

1. Logical Network Perimeter



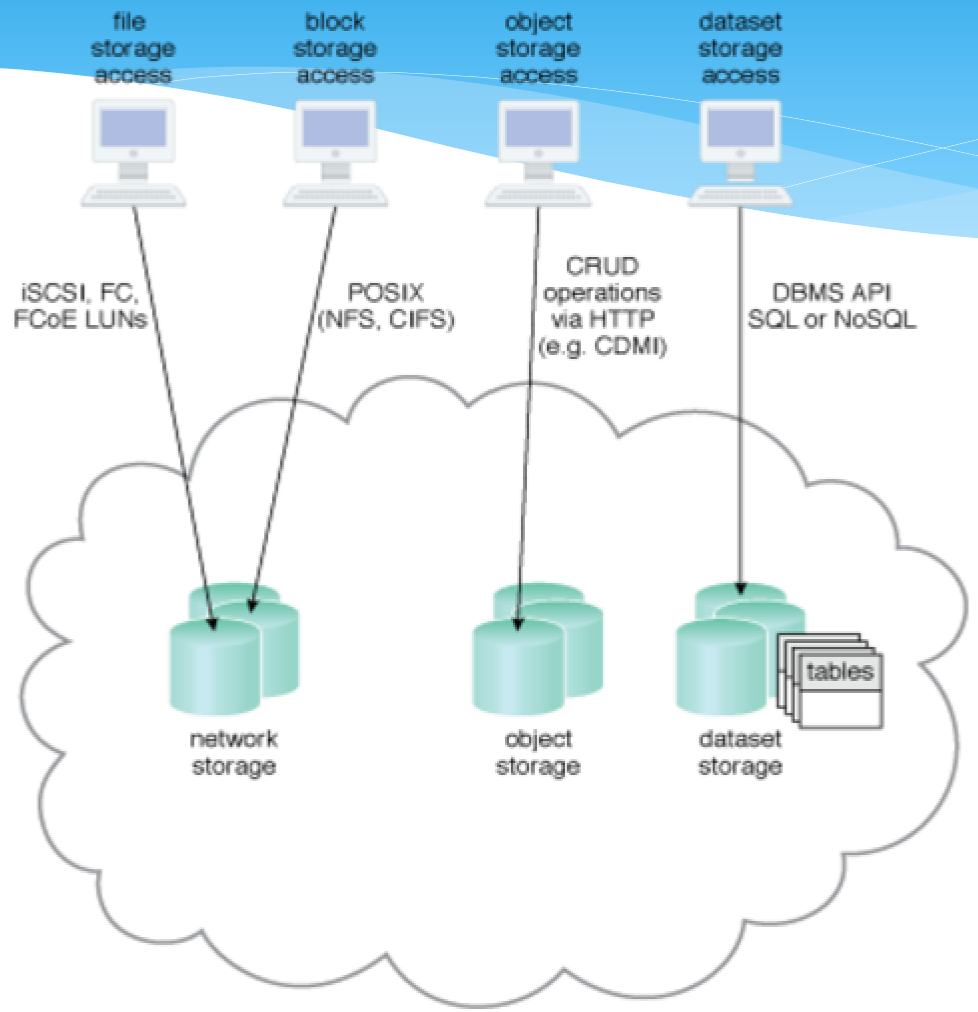
Two logical network perimeters surround the cloud consumer and cloud provider environments

2. Virtual Server



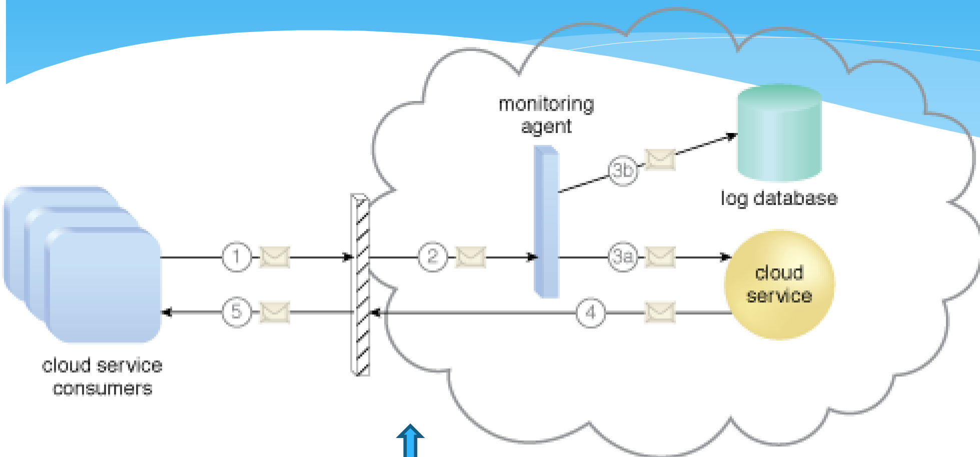
A virtual server hosts an active cloud service and is further accessed by a cloud consumer for administrative purposes

3. Cloud Storage Device

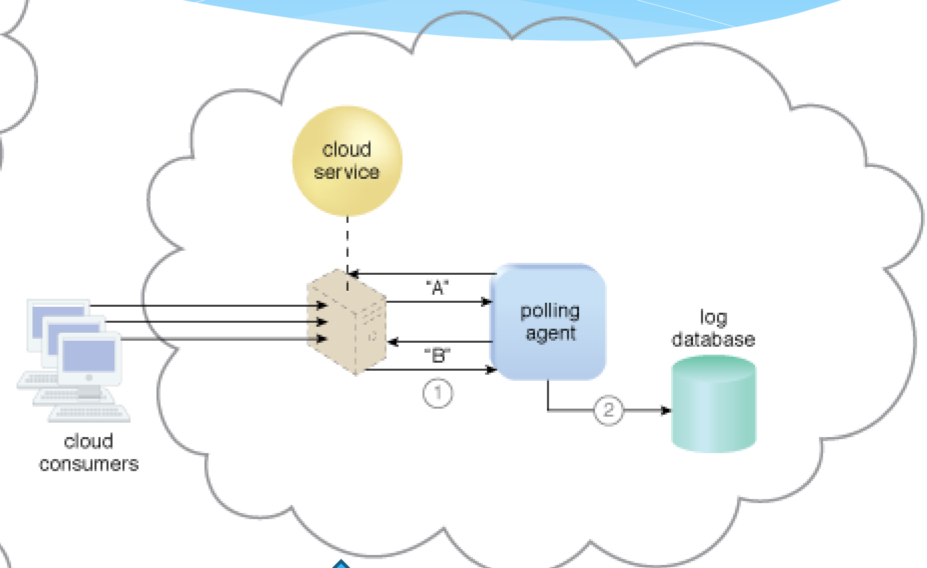


Different cloud service consumers utilize different technologies to interface with virtualized cloud storage devices

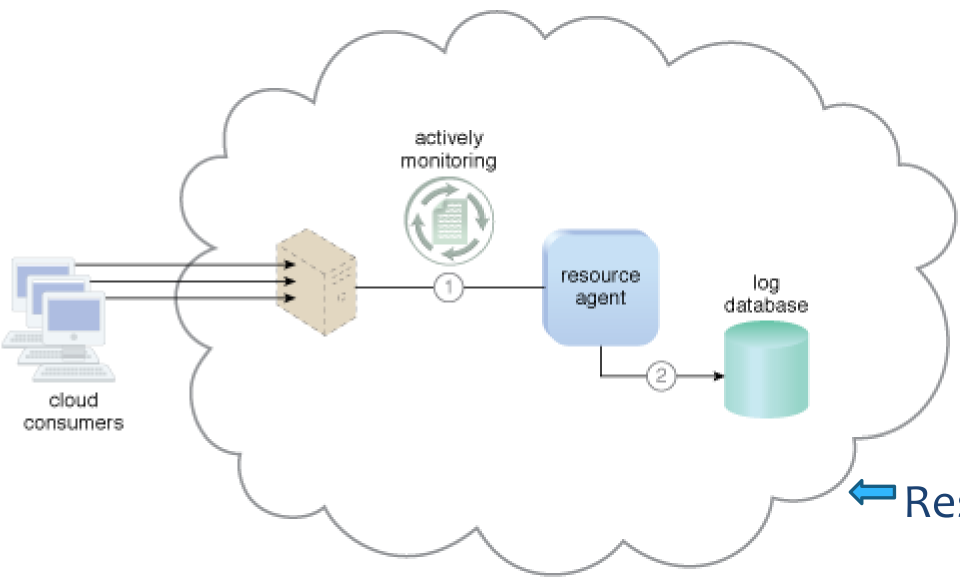
4. Cloud Usage Monitor



Monitoring Agent

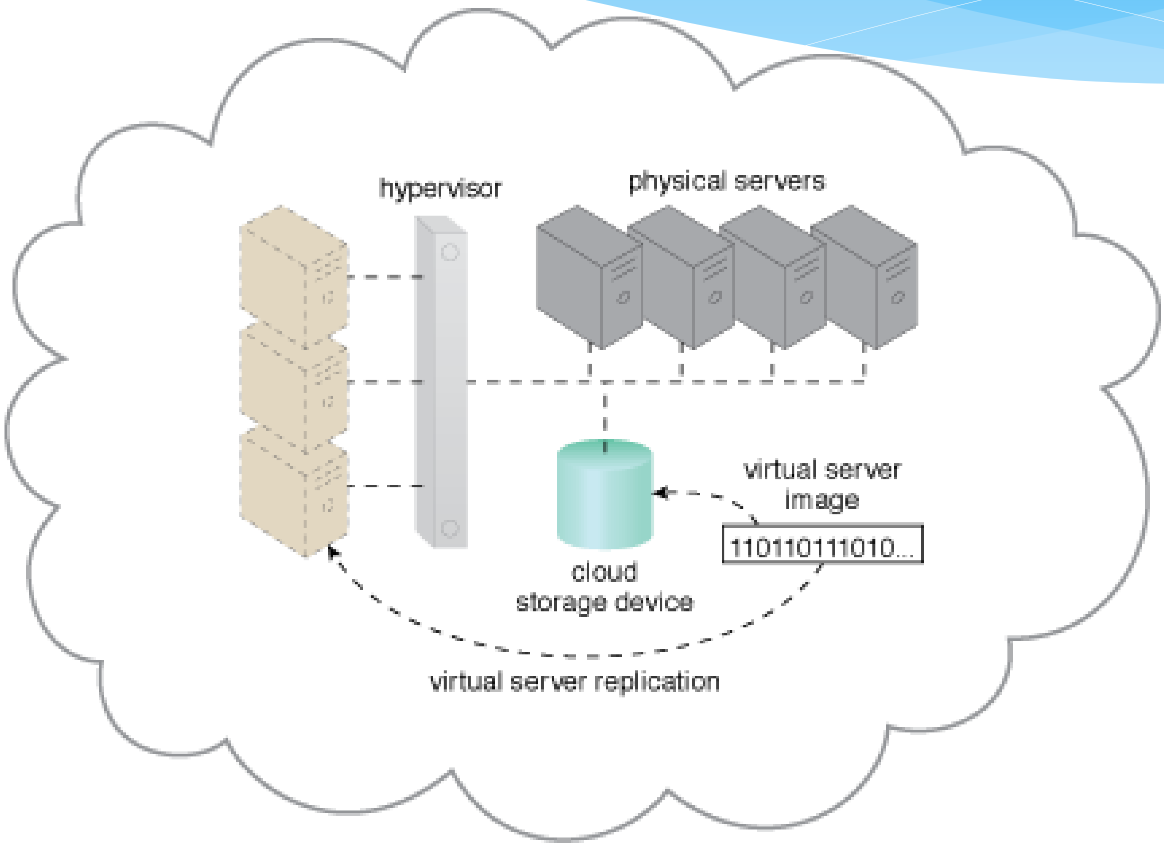


Polling Agent



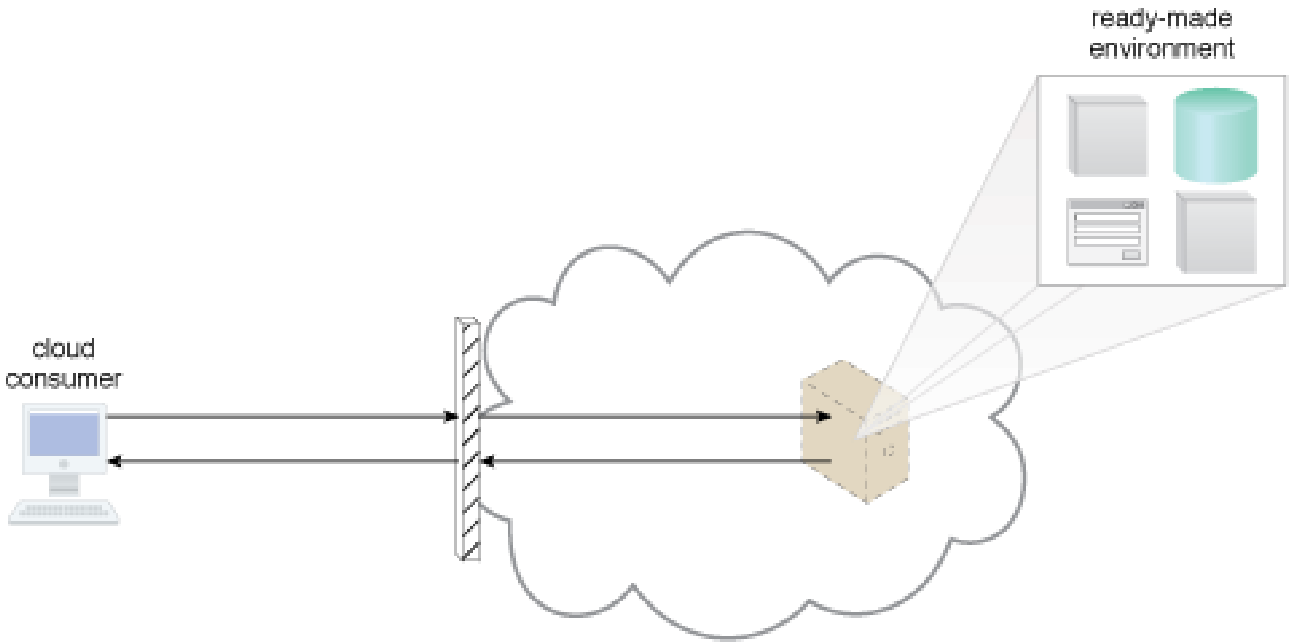
Resource Agent

5. Resource Replication



Hypervisor replicates several instances of a virtual server, using a stored virtual server image

6. Ready-made Environment



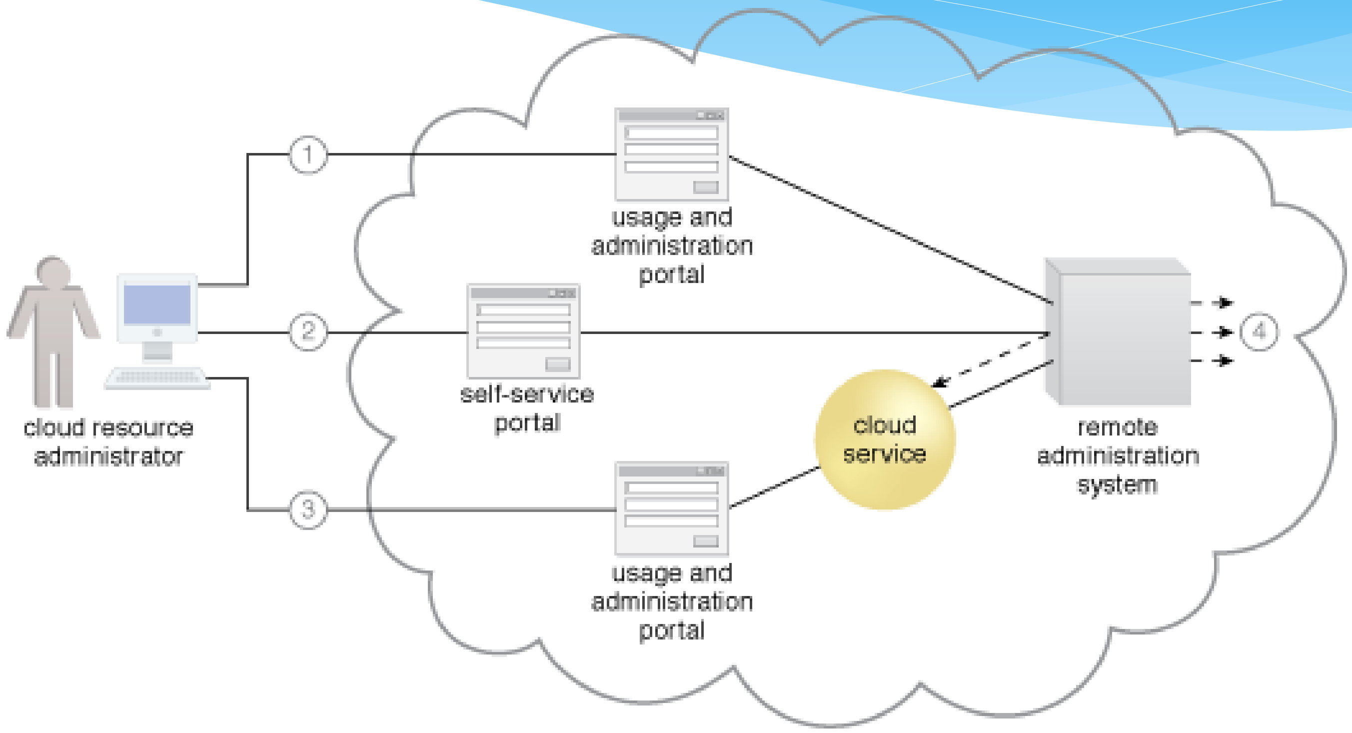
A cloud consumer accesses a ready-made environment hosted on a virtual server

Cloud Management Mechanisms

- Remote Administration System
- Resource Management System
- SLA Management System
- Billing Management System

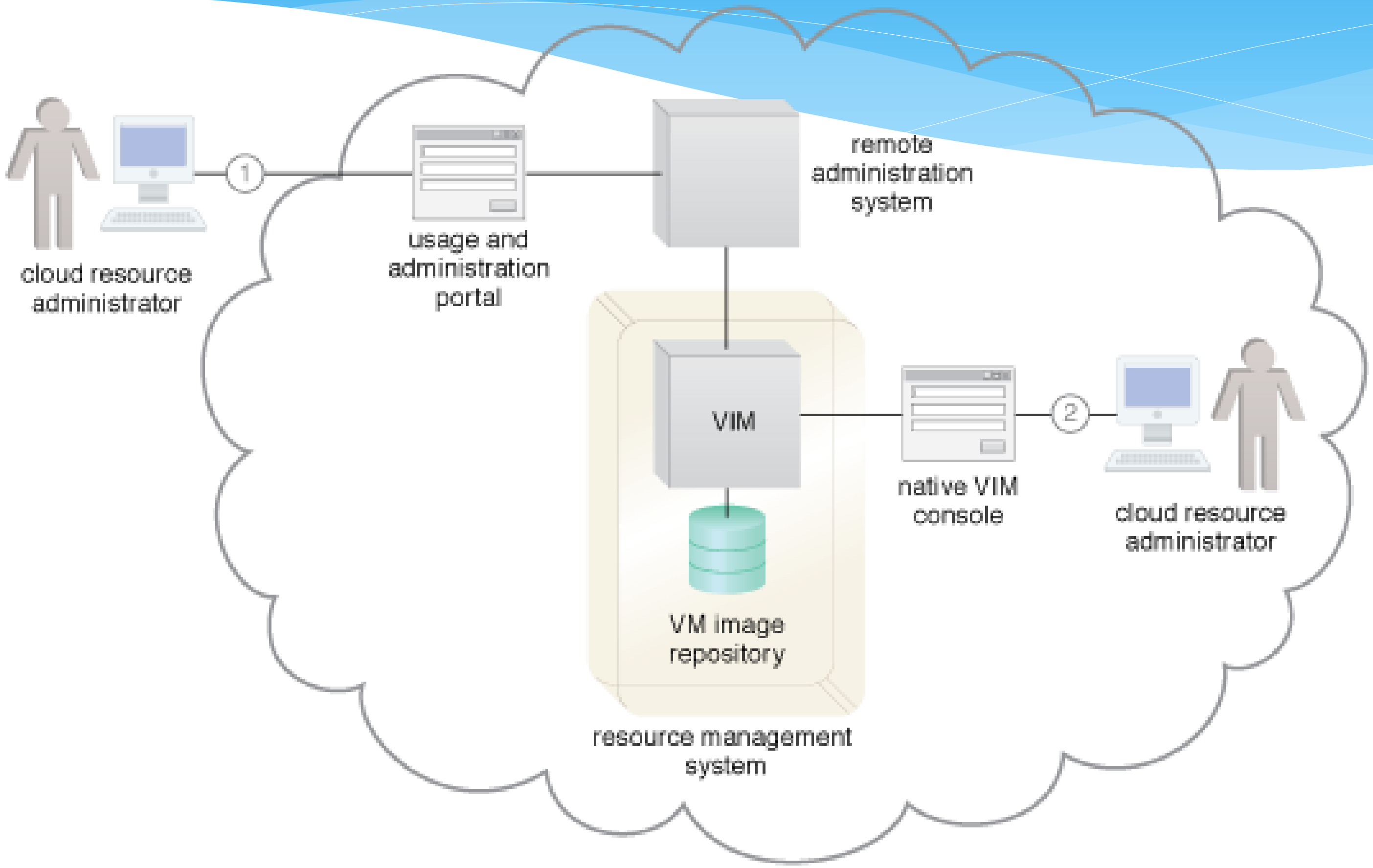
http://cloudpatterns.org/mechanisms/overview

1. Remote Administration System



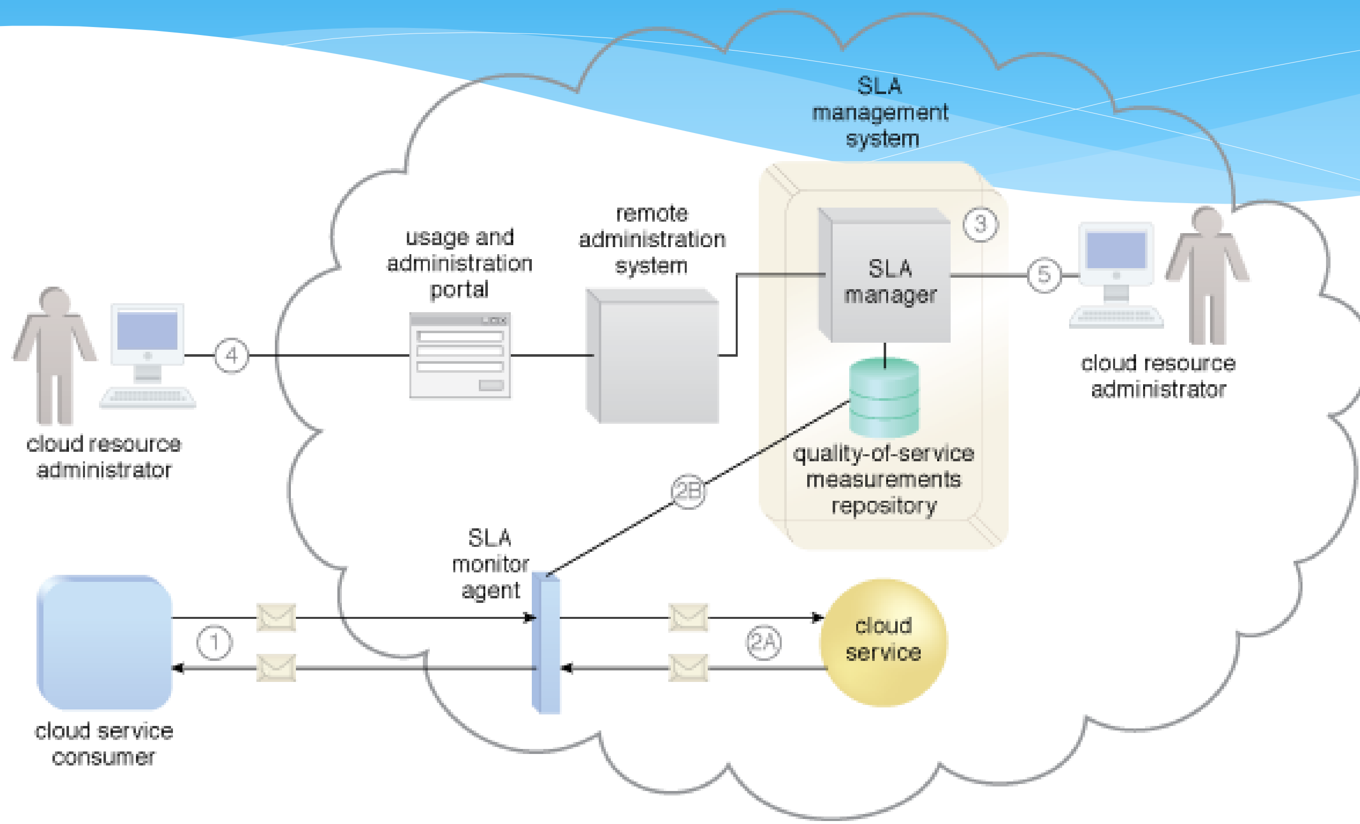
A cloud resource administrator uses the usage and administration portal to configure an already leased virtual server (not shown) to prepare it for hosting (1). The cloud resource administrator then uses the self-service portal to select and request the provisioning of a new cloud service (2). The cloud resource administrator then accesses the usage and administration portal again to configure the newly provisioned cloud service that is hosted on the virtual server (3). Throughout these steps, the remote administration system interacts with the necessary management systems to perform the requested actions (4).

2. Resource Management System



The cloud consumer's cloud resource administrator accesses a usage and administration portal externally to administer a leased IT resource (1). The cloud provider's cloud resource administrator uses the native user-interface provided by the VIM to perform internal resource management tasks (2).

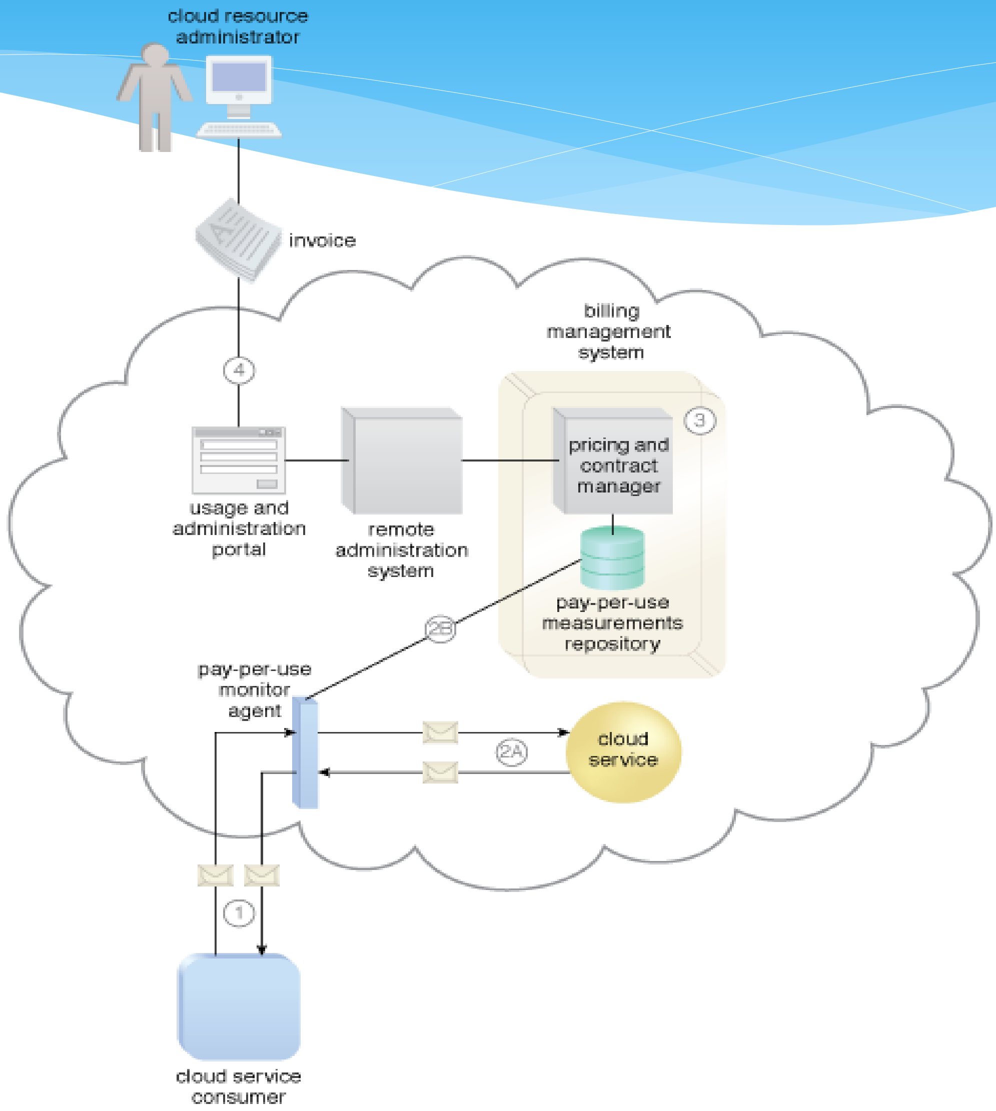
3. SLA Management System



The SLA monitor polls the cloud service by sending over polling request messages (MREQ1 to MREQN). The monitor receives polling response messages (M to M) that report that the service was "up" at each polling cycle (1a). The SLA monitor stores the "up" time—time period of all polling cycles 1 to N—in the log database (1b). The SLA monitor polls the cloud service that sends polling request messages (M to M). Polling response messages are not received (2a). The response messages continue to time out, so the SLA monitor stores the "down" time—time period of all polling cycles N+1 to N+M—in the log database (2b). The SLA monitor sends a polling request message (M) and receives the polling response message (M) (3a). The SLA monitor stores the "up" time in the log database (3b)

4. Billing Management System

A cloud service consumer exchanges messages with a cloud service (1). A pay-per-use monitor keeps track of the usage and collects data relevant to billing (2A), which is forwarded to a repository that is part of the billing management system (2B). The system periodically calculates the consolidated cloud service usage fees and generates an invoice for the cloud consumer (3). The invoice may be provided to the cloud consumer through the usage and administration portal (4).



Cloud Security Mechanisms

* Encryption
* Hashing
* Digital Signatures
* Public Key Infrastructure (PKI)
* Identity and Access Management (IAM)
* Single Sign-On (SSO)
* Cloud-Based Security Groups
* Hardened Virtual Server Images

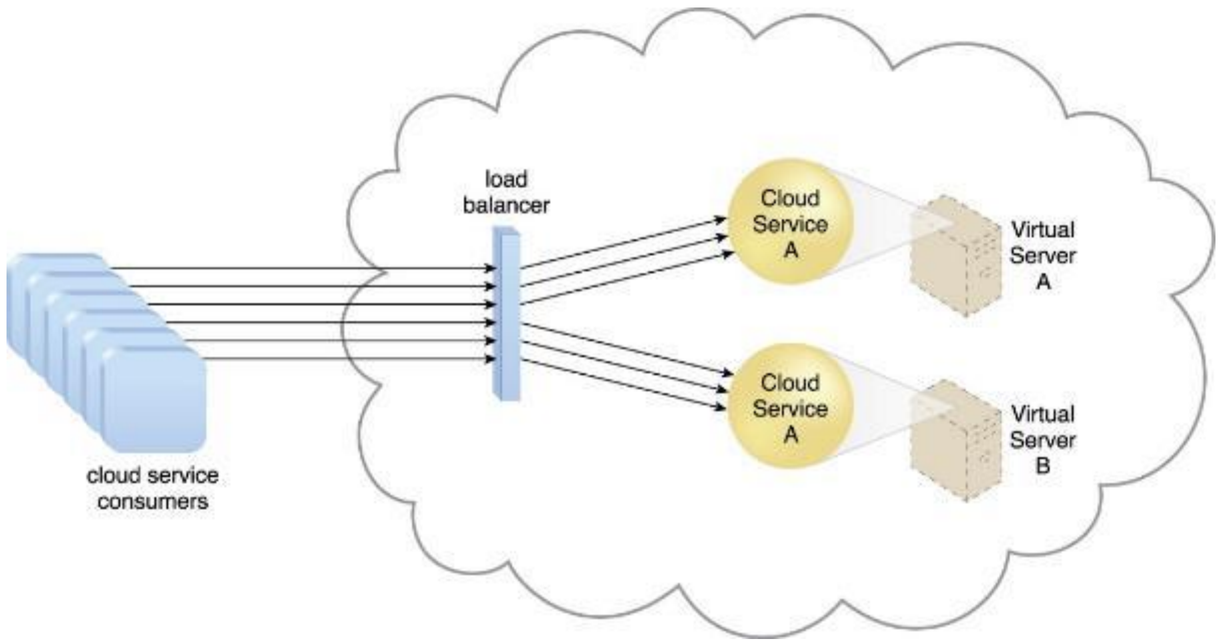
http://cloudpatterns.org/mechanisms/overview

Fundamental Cloud Architectures

* Workload Distribution Architecture
* Resource Pooling Architecture
* Dynamic Scalability Architecture
* Service Load Balancing Architecture

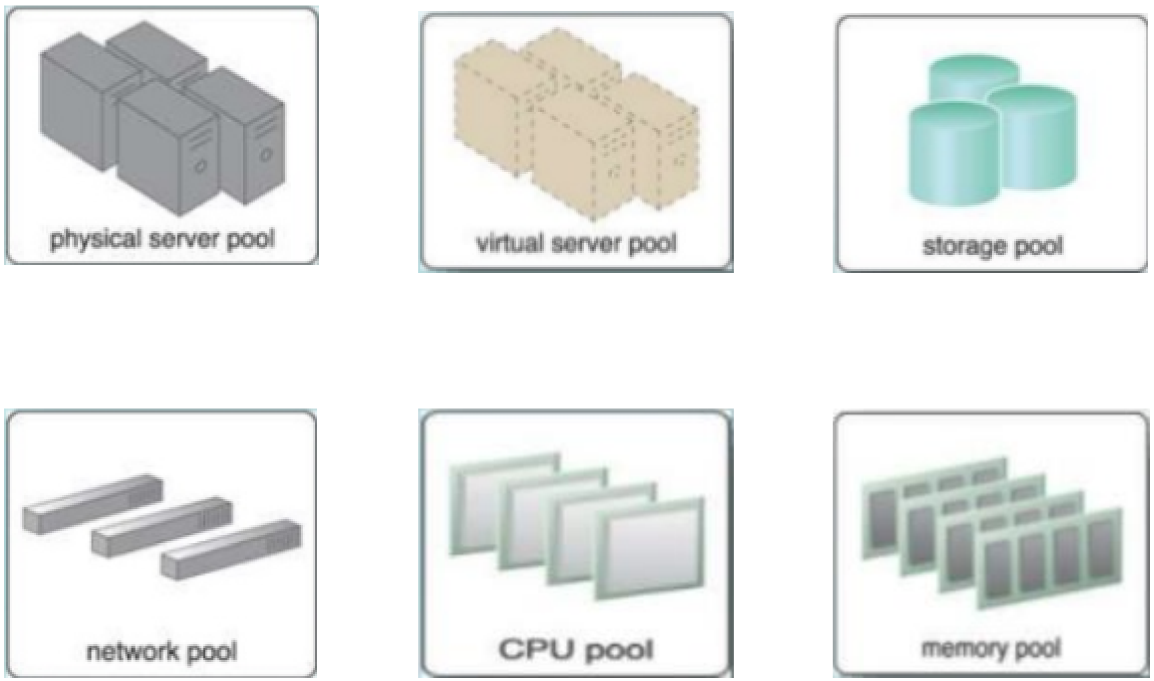
http://www.informit.com/articles/article.aspx?p=2093407

1. Workload Distribution Architecture

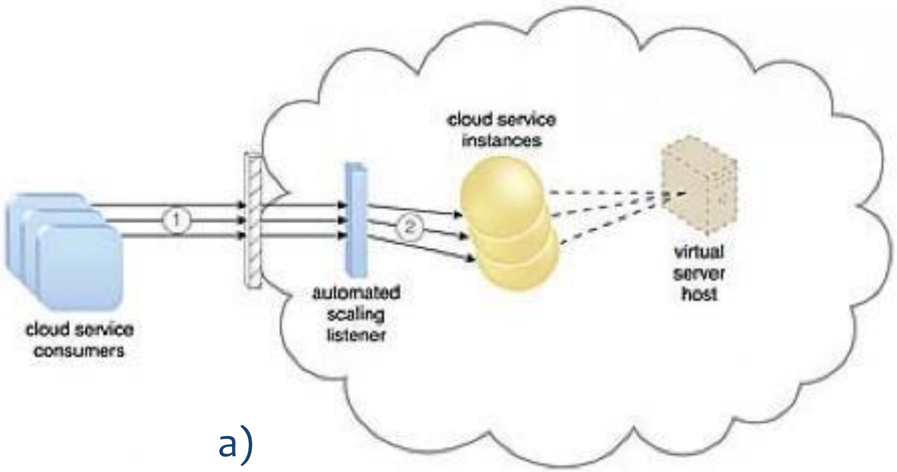


A redundant copy of Cloud Service A is implemented on Virtual Server B. The load balancer intercepts cloud service consumer requests and directs them to both Virtual Servers A and B to ensure even workload distribution.

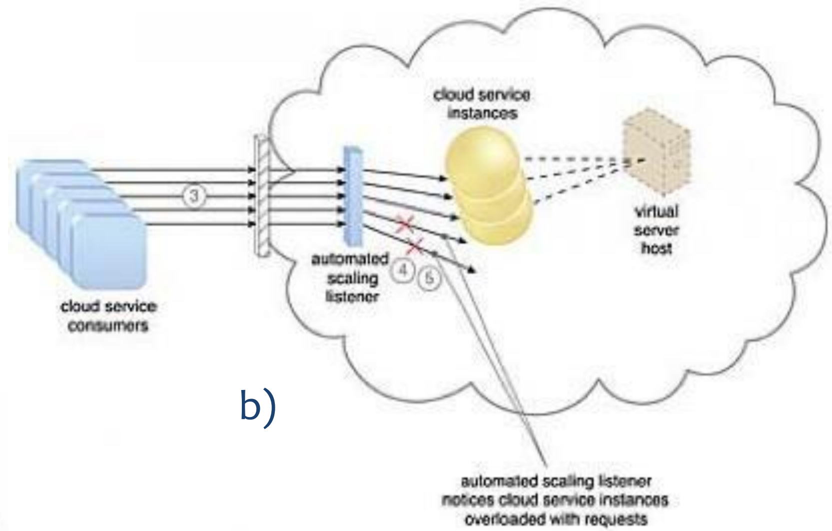
2. Resource Pooling Architecture



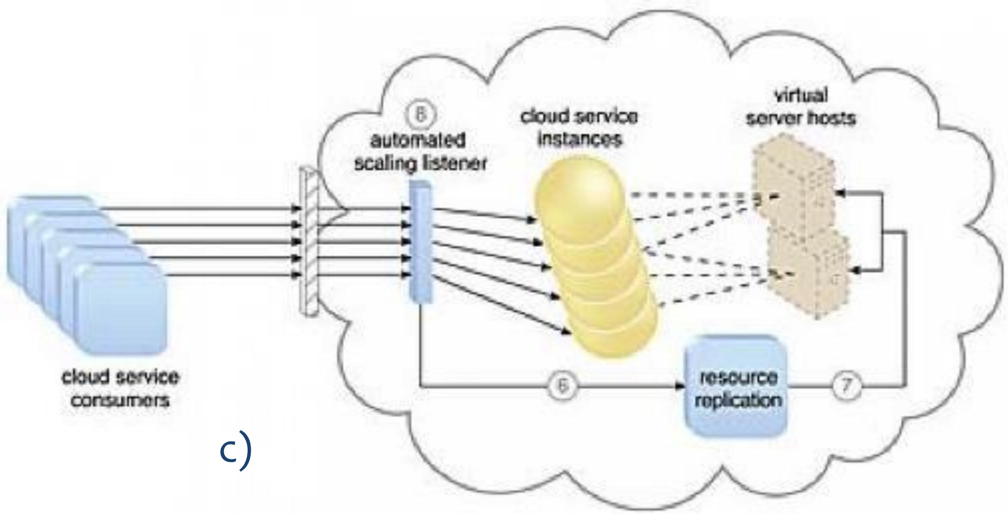
3. Dynamic Scalability Architecture



a)



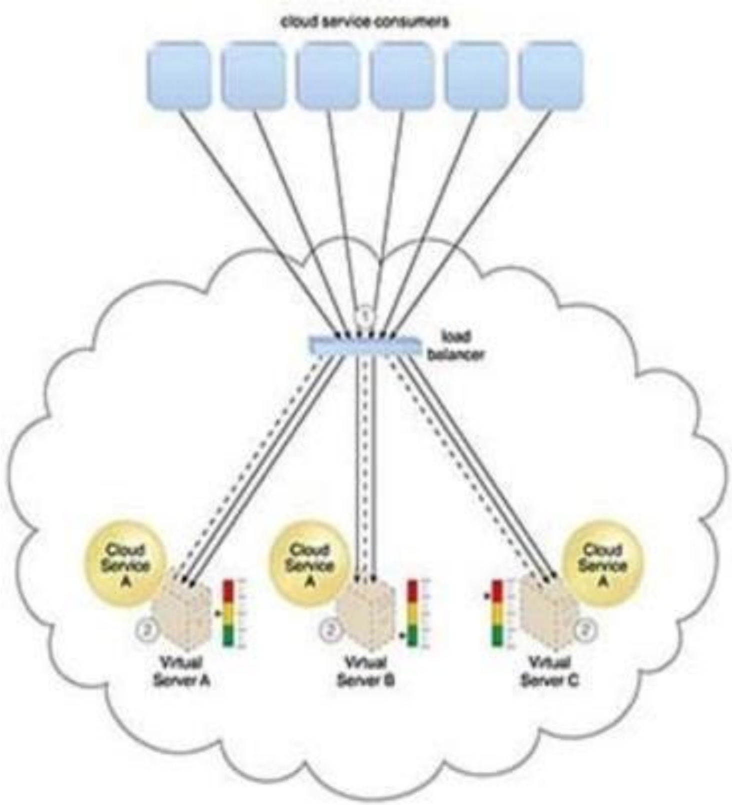
b)



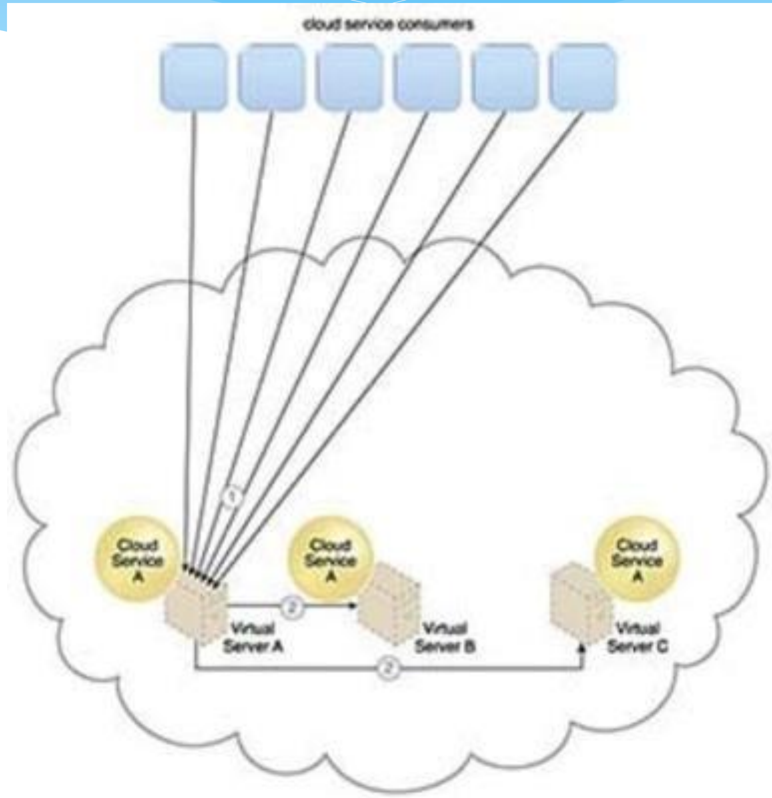
c)

The process of dynamic horizontal scaling

4. Service Load Balancing Architecture

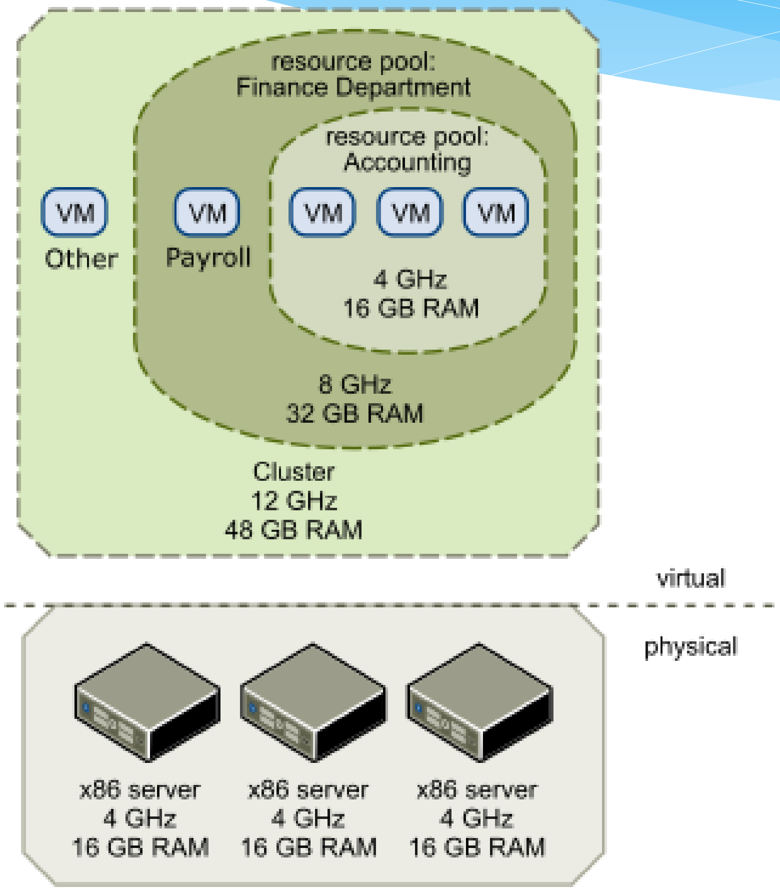


The load balancer intercepts messages sent by cloud service consumers (1) and forwards them to the virtual servers so that the workload processing is horizontally scaled (2).



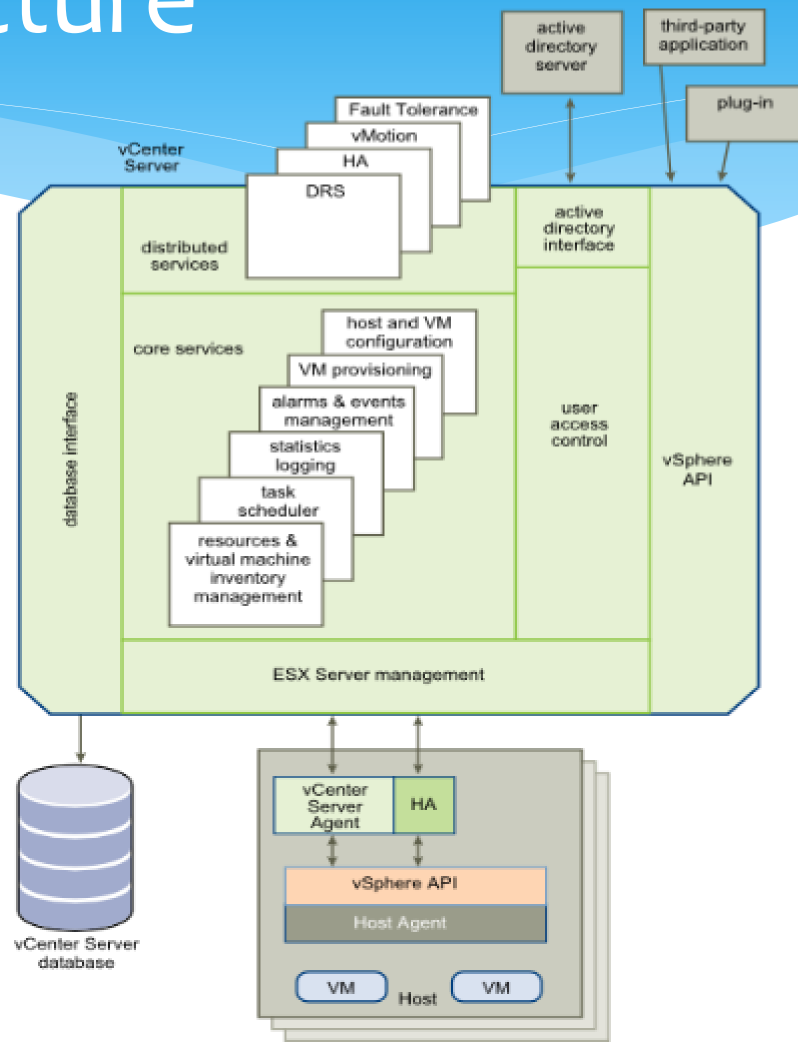
Cloud service consumer requests are sent to Cloud Service A on Virtual Server A (1). The cloud service implementation includes built-in load balancing logic that is capable of distributing requests to the neighboring Cloud Service A implementations on Virtual Servers B and C (2).

Hosts, Clusters and Resource Pools

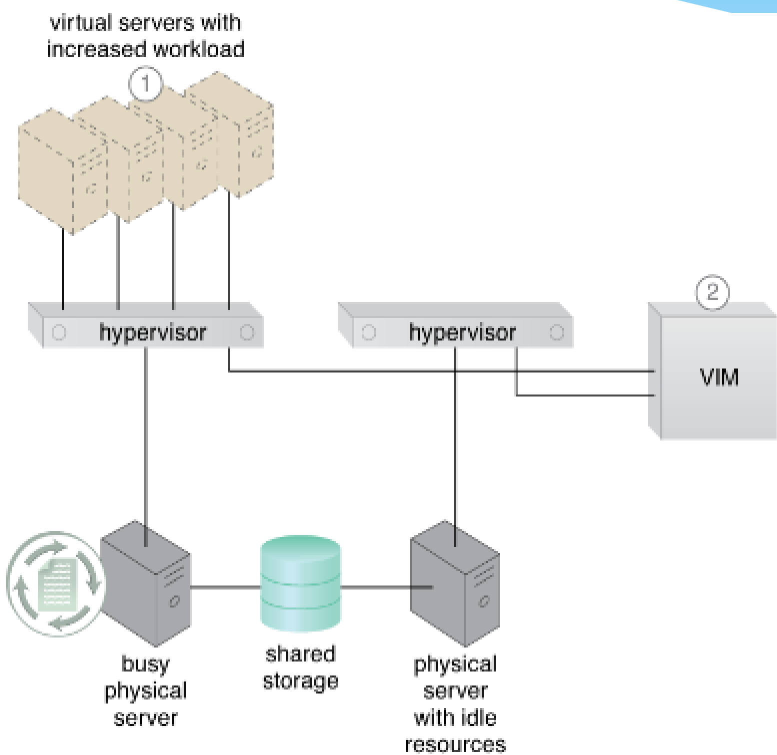


VCenter Management Server Architecture

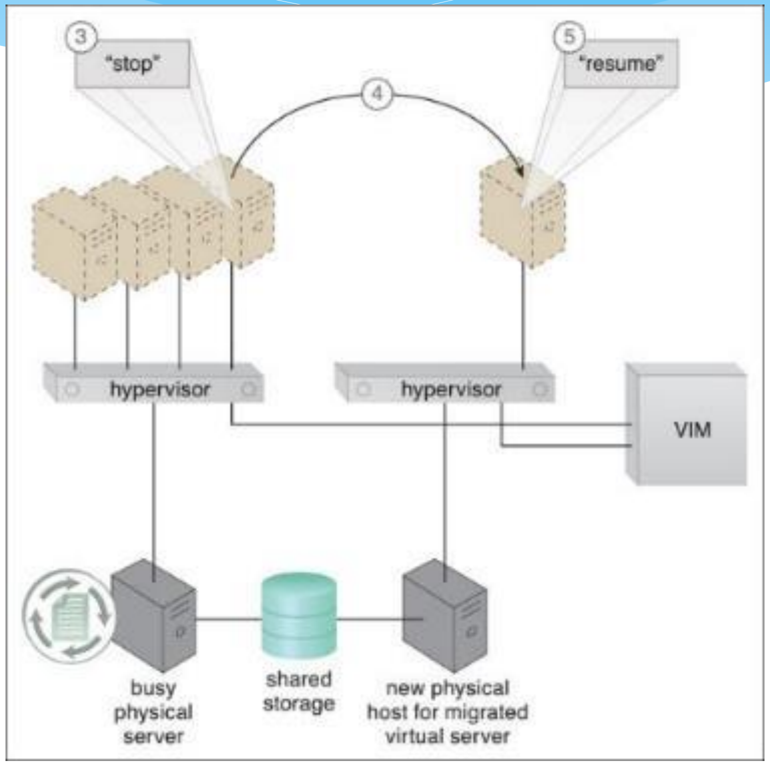
* VCenter Server Components:
  * User Access Control
  * Core Services
  * Distributed services
  * Plug-ins
  * Interfaces



VMotion

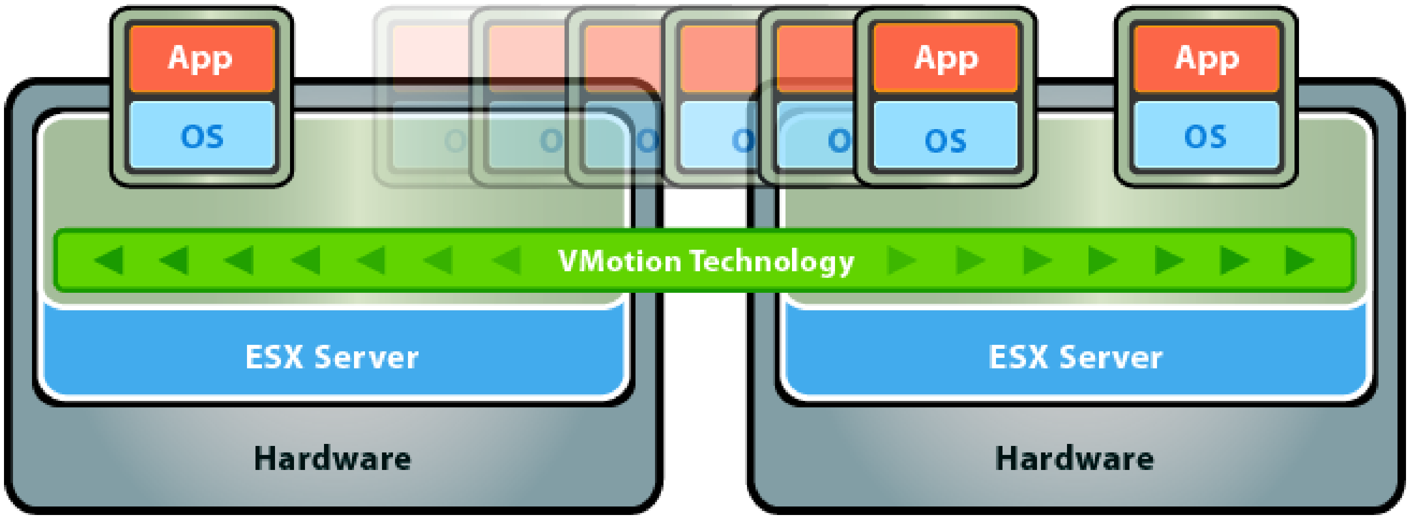


A virtual server capable of auto-scaling experiences an increase in its workload (1). The VIM decides that the virtual server cannot scale up because its underlying physical server host is being used by other virtual servers (2).



The VIM commands the hypervisor on the busy physical server to suspend execution of the virtual server (3). The VIM then commands the instantiation of the virtual server on the idle physical server. State information (such as dirty memory pages and processor registers) is synchronized (4). The VIM commands the hypervisor at the new physical server to resume the virtual server processing (5).

VMotion



VMware VMotion moves live, running virtual machines from one host to another while maintaining continuous service availability.

https://www.vmware.com/pdf/vmotion_datasheet.pdf